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Cross-Slice Transformer with Adaptive Slice Fusion for Lymph Node Segmentation in CT Imaging

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ABSTRACT Accurate lymph node assessment on computed tomography (CT) is essential for cancer staging and treatment planning, yet automation remains challenging due to small node size, low soft-tissue contrast, and extreme class imbalance (foreground voxels comprising below 1% of the volume). We propose the Cross-Slice Transformer (CST), a lightweight 2.5D architecture that learns adaptive slice importance through statistical descriptor extraction and transformer-based attention over five consecutive CT slices. CST is evaluated against three 2.5D baselines (U-Net, TransUNet, SwinUNet) and the 3D nnUNet framework on two publicly available TCIA datasets: a multi-region dataset (Ver1, 296 patients, mediastinal and abdominal) and a mediastinal-focused dataset (Ver2, 206 patients). While nnUNet-3D serves as a strong 3D reference (0.589 Dice on Ver2), the central research questions concern 2.5D architectural design and dataset composition, which a single 3D model cannot address. On Ver2, CST achieves the highest Dice among 2.5D models (0.543 ± 0.044), with the highest recall among 2.5D models (0.618 ± 0.024) and a compact footprint of $7.97\,$M parameters, substantially fewer than the 2.5D baselines (25–99 M). Node-level detection analysis reveals that CST leads all models including nnUNet-3D on detection rate across all size strata on Ver2: 53.6% for small nodes ($<10\,$mm), 82.9% for medium (10–15 mm), and 92.5% for large ($\geq 15\,$mm) at IoU ≥ 0.1. All models improve consistently when trained on anatomically focused single-region data, with Dice gains of 0.047–0.152 points from Ver1 to Ver2, demonstrating that dataset composition is a primary driver of segmentation performance across all architectural families. Ablation experiments confirm the contribution of each CST component: adaptive transformer attention ($+0.006$ Dice over fixed uniform weights), the interaction descriptor ($+0.008$ over a reduced descriptor), and per-case intensity normalisation ($+0.008$ over fixed HU windowing). These results provide evidence-based guidance for 2.5D architecture selection and dataset curation in clinical lymph node segmentation.

INDEX TERMS Cross-Slice Transformer, lymph node segmentation, CT imaging, adaptive slice fusion, 2.5D processing, deep learning, transformer attention, anatomy-guided segmentation

I. INTRODUCTION

LYMPH node metastasis remains one of the strongest prognostic indicators in oncology and directly affects staging, treatment planning, and overall survival across almost every solid malignancy. In the TNM staging system, nodal status carries equal weight to the primary tumour and distant metastases, illustrating how crucial accurate nodal assessment is in daily clinical practice. The presence, number, and location of metastatic nodes can completely alter the therapeutic approach and provide critical information about patient prognosis [1]. In breast cancer, the degree of axillary involvement determines whether adjuvant chemotherapy is indicated and how aggressive it should be [2]. In lung cancer, metastatic mediastinal nodes typically contraindicate surgery and redirect patients toward chemoradiation [3]. In gastroesophageal cancers, nodal burden guides neoadjuvant therapy and the extent of resection [4]. Because these decisions carry major consequences for morbidity, quality of life, and healthcare costs, reliable lymph node assessment is an essential component of modern oncology.

A. LIMITATIONS OF CURRENT CLINICAL PRACTICE

Computed tomography remains the primary modality for lymph node evaluation in routine oncologic practice due to its widespread availability, rapid acquisition, and whole-body coverage. Current radiological assessment relies primarily on size criteria: nodes exceeding 10 mm in short-axis diameter are considered suspicious according to RECIST 1.1 guidelines [5]. This size-based approach has well-recognised limitations, with reported sensitivities of 57–76% and specificities of 55–82% across cancer types. The problem is bidirectional: micrometastases frequently occur in normal-sized nodes (false negatives), while reactive enlargement causes benign nodes to exceed the threshold (false positives). Manual identification across hundreds of CT slices is time-consuming and shows considerable inter-observer variability (15–35%) even among experienced radiologists [6].

B. TECHNICAL CHALLENGES IN AUTOMATED ANALYSIS

Automated lymph node segmentation faces several distinct difficulties. Lymph nodes are small (typically 3–30 mm), placing them near the resolution limit of clinical CT scanners, with Hounsfield units similar to adjacent muscle and vessels. They occupy less than 1% of voxels in whole-body scans, creating severe class imbalance that biases standard algorithms toward the background. Substantial inter-patient variability in node size, shape, and anatomical distribution further complicates generalisation, and morphological similarity to vessels and muscle bundles introduces persistent segmentation uncertainty even for expert readers.

C. DEEP LEARNING FOR MEDICAL IMAGE SEGMENTATION

Over the past decade, deep learning has driven major progress in medical image segmentation. U-Net [7] established the standard encoder–decoder architecture with skip connections, enabling precise spatial localisation while preserving multi-scale contextual information. ResNet [8] demonstrated that residual connections facilitate training of very deep networks by mitigating vanishing-gradient issues. Transformer-based architectures have more recently been adapted for medical imaging. TransUNet [9] integrates a convolutional encoder with transformer blocks to combine fine-grained local features with long-range spatial dependencies. SwinUNet [10] adopts a hierarchical Swin Transformer [11] backbone for both encoding and decoding, demonstrating that transformer-dominant architectures can achieve competitive segmentation performance when tailored to medical imaging tasks. nnUNet [12] introduced a self-configuring 3D framework that automatically adapts preprocessing and network topology to any target dataset, and consistently achieves state-of-the-art results across diverse segmentation benchmarks. Attention mechanisms have shown benefit across cancer imaging modalities. Recent work in mammography demonstrates that multi-scale feature extraction and specialised loss functions effectively address small target size and severe class imbalance [13], suggesting similar principles are applicable to CT-based lymph node segmentation. For volumetric CT data, full 3D convolutions capture rich spatial context but quickly exhaust GPU memory on clinical scan sizes. A practical alternative is the 2.5D strategy, which processes several adjacent slices simultaneously, balancing volumetric awareness with computational efficiency [14], [15]. Cross-slice attention mechanisms provide a means of exchanging contextual information between neighbouring slices without the overhead of full 3D processing [16]. TotalSegmentator [17] enables fully automated organ localisation from CT without manual annotation, making anatomy-guided region-of-interest extraction practical at scale.

D. CONTRIBUTIONS AND PAPER ORGANISATION

Most existing lymph node segmentation studies evaluate a single architecture on a single dataset configuration, leaving two important questions unanswered. First, how does anatomical dataset composition systematically affect segmentation performance across diverse architectural families? Second, which 2.5D architectural design choices most benefit node-level detection in memory-constrained deployment scenarios where full 3D processing is not feasible? nnUNet-3D serves as a strong 3D reference in our evaluation, but being a single model it cannot answer either question. This work addresses these gaps through the following contributions:

1) **Cross-Slice Transformer (CST).** We propose a lightweight 2.5D architecture that learns adaptive slice importance by combining statistical descriptors with transformer-based self-attention across five consecutive CT slices, enabling context-aware adaptive slice fusion. CST is integrated with a U-Net decoder for end-to-end segmentation and has 7.97 M parameters, substantially fewer than the 2.5D baselines (25–99 M), making it suitable for memory-constrained clinical deployment.
2) **Anatomy-guided preprocessing pipeline.** We introduce a fully automated corridor-crop pipeline using TotalSegmentator organ masks for patient-specific region-of-interest extraction and an anatomy prior channel that encodes the spatial neighbourhood of mediastinal lymph node stations without manual annotation.
3) **Systematic dual-dataset evaluation.** We evaluate all five models on two datasets with contrasting anatomical scope (multi-region Ver1 and mediastinal-focused Ver2) under a unified protocol matching nnUNet cross-validation splits, enabling direct comparison of how dataset composition drives performance across architectural families.
4) **Comprehensive clinical evaluation framework.** Beyond aggregate Dice, we report size-stratified node detection rates using RECIST short-axis strata and an IoU ≥ 0.1 detection threshold appropriate for small structures, providing clinically actionable performance characterisation across node sizes.

The remainder of the paper is organised as follows. Section II reviews related work. Section III describes datasets, preprocessing, architectures, and training. Section IV presents experimental results and ablation study. Section V discusses findings and limitations. Section VI concludes with future directions.

II. RELATED WORK

A. U-NET BASED ARCHITECTURES

U-Net and its variants remain the most widely used backbone for lymph node segmentation. Manjunatha et al. [18] showed that a modified U-Net with residual learning improved lymph node detection in CT scans by effectively handling small nodes in both abdominal and mediastinal regions. Al Hasan et al. [19] proposed a two-stage deep learning framework for automated segmentation of lymph nodes on neck CT scans, achieving high sensitivity while reducing false positives through 3D CNN refinement.

B. TRANSFORMER-BASED ARCHITECTURES

Hybrid CNN–transformer models have gained increasing attention in medical image segmentation. TransUNet [9] employed a ResNet-50 backbone for feature extraction followed by transformer layers embedded within a U-Net-style decoder, achieving 77.48% average Dice on the Synapse multi-organ dataset. Swin Transformer [11] introduced shifted-window self-attention and hierarchical feature representations to improve computational efficiency while capturing long-range dependencies. Building on this foundation, SwinUNet [10] applied a transformer-dominant architecture to both encoder and decoder stages, demonstrating strong segmentation performance across medical imaging benchmarks. For lymph node-specific applications, Yu et al. [20] addressed positional bias in transformer-based models by introducing location-debiased query selection and contrastive query representations.

C. 3D FULLY AUTOMATED PIPELINES

nnUNet [12] introduced a self-configuring segmentation framework that automatically adapted preprocessing, network topology, and training hyperparameters to any target dataset. Without task-specific modifications, it consistently achieved competitive or state-of-the-art results across diverse medical segmentation benchmarks, and is widely used as a strong reference baseline in the field.

D. 2.5D PROCESSING AND ANATOMY-GUIDED STRATEGIES

Because full 3D convolutions quickly exhaust GPU memory on clinical CT volumes, most practical systems rely on 2.5D approaches that feed several neighbouring slices simultaneously. Roth et al. [15] trained deep CNNs on randomly oriented 2.5D patches for lymph node detection. Wang et al. [16] enriched ordinary 2D networks by learning explicit contextual embeddings from adjacent slices. Kumar et al. [14] developed a flexible 2.5D segmentation approach with in-slice and cross-slice attention, demonstrating effective inter-slice modelling without the memory overhead of full 3D processing. TotalSegmentator [17] provided fully automated segmentation of 104 anatomical structures from CT without manual annotation, enabling robust patient-specific region-of-interest extraction for downstream tasks.

E. LOSS FUNCTIONS AND ANNOTATION QUALITY

Extreme foreground-background imbalance is a persistent issue in lymph node segmentation. Tversky Loss [21] generalised the Dice coefficient by introducing independent weights for false positives and false negatives, allowing the loss to prioritise recall when missed detections carry higher clinical cost than false alarms. Soft Dice Loss [22] optimised the Dice overlap directly in a differentiable form, providing stable gradient signal for foreground regions regardless of their size relative to the background. Binary cross-entropy remained a common auxiliary term that supplied dense per-voxel supervision and stabilised training in early epochs before overlap-based terms became informative [12]. Composite losses combining these terms showed consistent improvements over any single term alone on small-structure segmentation tasks with severe class imbalance [21], [22].

Annotation quality compounds the class-imbalance problem in practice. Song et al. [23] investigated learning from AI-generated annotations that contain systematic errors, proposing a confidence-regularised co-teaching algorithm that substantially mitigated annotation noise by training two models simultaneously and updating parameters from disagreement-confident examples. Their findings are directly relevant to partially annotated medical datasets where labels cover only a subset of the volume: unlabelled regions introduce implicit background supervision that penalises correct predictions outside the annotated band. The TCIA collections used in this work include such partially annotated patients; Section III-A describes the strategies evaluated for handling them and the rationale for the final exclusion decision.

III. METHODOLOGY

A. DATASETS

We evaluate on two publicly available CT lymph node datasets sourced from the TCIA repository, differentiated primarily by anatomical scope.

1) Dataset Ver1 — Multi-Region (Mediastinal and Abdominal)

Ver1 comprises **296 fully annotated patients** covering both mediastinal and abdominal lymph node regions, drawn from the *CT Lymph Nodes* [24] and *Mediastinal-Lymph-Node-SEG* [25] TCIA collections. The source collections include partially annotated patients whose labels cover only a narrow band of slices around confirmed nodes, leaving the majority of the volume unlabelled. Three strategies were evaluated for handling this subset. Including partially annotated patients in training alongside fully annotated data degraded

Dice, as the model was penalised for correct predictions on unlabelled slices treated implicitly as background. A two-stage curriculum — pre-training on fully annotated patients followed by fine-tuning with partial annotations — similarly failed to improve over the fully-annotated-only baseline. A third approach, excluding partial annotations from validation while retaining them in training, produced the same degradation. Given these findings, partially annotated patients were excluded from all experiments. After exclusion, Ver1 contains **1,770 annotated lymph node instances** with a mean of 6.0 nodes per patient. Lymph node short-axis diameter ranges from 0.5 to 82.5 mm (mean 14.9 ± 10.7 mm); 65% of nodes exceed 10 mm. Mean CT volume depth is 243 ± 99 slices; positive slices account for $25.4 \pm 16.6\%$ of each volume. Class imbalance is severe: median foreground fraction 0.24% (BG:FG ratio $414:1$, IQR 186–1,243).

2) Dataset Ver2 — Mediastinal Only

Ver2 is the mediastinal subset of Ver1, restricted to mediastinal patients from both TCIA collections [24], [25], comprising **206 usable fully annotated patients**. The 206 patients contain **1,250 lymph node instances** (mean 6.1 ± 4.5 per patient). Node short-axis diameter spans 0.5–65.0 mm (mean 16.9 ± 10.5 mm); 57% of nodes exceed 10 mm. Positive slices account for $20.4 \pm 15.1\%$ per volume; median foreground fraction 0.35% (BG:FG $286:1$, IQR 135–582).

Dataset overlap. Ver2 is a strict subset of Ver1; they are not independent cohorts. Any performance differences between Ver1 and Ver2 therefore reflect both anatomical scope and dataset size simultaneously, rather than either factor alone. Results are reported as separate benchmarks to characterise each configuration independently, and this confound is discussed further in Section V.

Dataset statistics are summarised in Table 1. Five-fold cross-validation was adopted to match the nnUNet auto-configuration protocol [12], enabling a directly comparable evaluation. All five models use identical fold assignments: **236 training and 60 validation patients per fold for Ver1**, and **165 training and 41 validation patients per fold for Ver2**. All reported Dice scores reflect identical patient-level test sets across all models.

B. PREPROCESSING PIPELINE

The preprocessing converts raw NIfTI CT volumes into standardised, corridor-cropped arrays shared identically by both CST and nnUNet. Fig. 1 illustrates the full pipeline; all steps are implemented with SimpleITK and TotalSegmentator [17].

1) Step 1 — Automatic Organ Segmentation

TotalSegmentator [17] (task `total`, fast mode) provides 34 anatomical organ masks per patient without manual annotation. Masks used downstream: aorta, heart, trachea, oesophagus, superior vena cava (SVC), inferior vena cava (IVC), and pulmonary artery (PA) (anatomy prior); bilateral lung lobes and vertebrae (VOI bounding and corridor anchor for Ver2). **Quality assurance.** TotalSegmentator achieves a mean Dice of 0.943 across structures on 1,200+ multi-institution CT volumes [17]. Two safeguards mitigate the risk of silent anchor corruption: (i) the corridor anchor uses the *median* organ centroid across all Z-slices, making it robust to isolated slice failures; and (ii) a 35 mm hard safety zone prevents the corridor from intersecting the aorta or heart. Visual inspection of all 206 Ver2 patients found no misaligned cases.

2) Step 2 — VOI Extraction and Resampling

A patient-specific VOI is extracted per patient. For Ver2, the Z-extent is bounded by the union of all lung lobe masks, capturing the full mediastinum from apex to base. For Ver1, the Z-extent spans the full body to include both thoracic and abdominal regions. All VOIs are resampled to 1.5 mm axial $\times$ 0.5 $\times$ 0.5 mm in-plane using B-spline interpolation for CT and nearest-neighbour for masks. Source volumes have native axial spacing of 1.0–5.0 mm (upsampling up to $3\times$); we observe no systematic correlation between native slice thickness and per-patient Dice in our validation set.

3) Step 3 — Anatomy Prior Mask

An anatomy prior binary mask guides the model toward regions where lymph nodes typically reside, defined as the union of seven central structures dilated by r mm:

$$M_{\text{prior}} = \bigcup_{k \in \mathcal{S}} \left(M_k \oplus B_r \right), \tag{1}$$

where $\mathcal{S} = \{$aorta, heart, trachea, oesophagus, SVC, IVC, PA$\}$, M_k is the TotalSegmentator mask for structure k, and B_r is a spherical structuring element of radius r. The submitted model uses $r = 5$ mm (tight prior, $\approx$22% corridor coverage). The effect of dilation radius on segmentation performance is evaluated in the ablation study (Section IV-D).

4) Step 4 — Corridor Crop

All volumes are cropped to a standardised **corridor**: 256×256 px (128×128 mm) for Ver2 and 384×384 px (192×192 mm) for Ver1. **Ver2 — spine-anchored.** The median vertebral centroid is shifted 30 mm anteriorly to centre the corridor over the anterior mediastinum. A 35 mm safety zone around the aorta and heart prevents corridor intersection with these structures. **Ver1 — aorta-anchored.** The median aorta centroid serves as the corridor centre, with a 35 mm safety zone around the aorta and IVC. **Width justification.** The 256 px width retains 90.1% of in-VOI lymph node annotations inside the corridor boundary; the remaining 9.9% are lateral nodes that partially fall outside the 128 mm FOV. The corridor is applied identically to all models.

5) Step 5 — Intensity Normalisation

Per-patient Z-score normalisation is applied at runtime inside the DataLoader. Statistics μ_α and σ_α are computed from

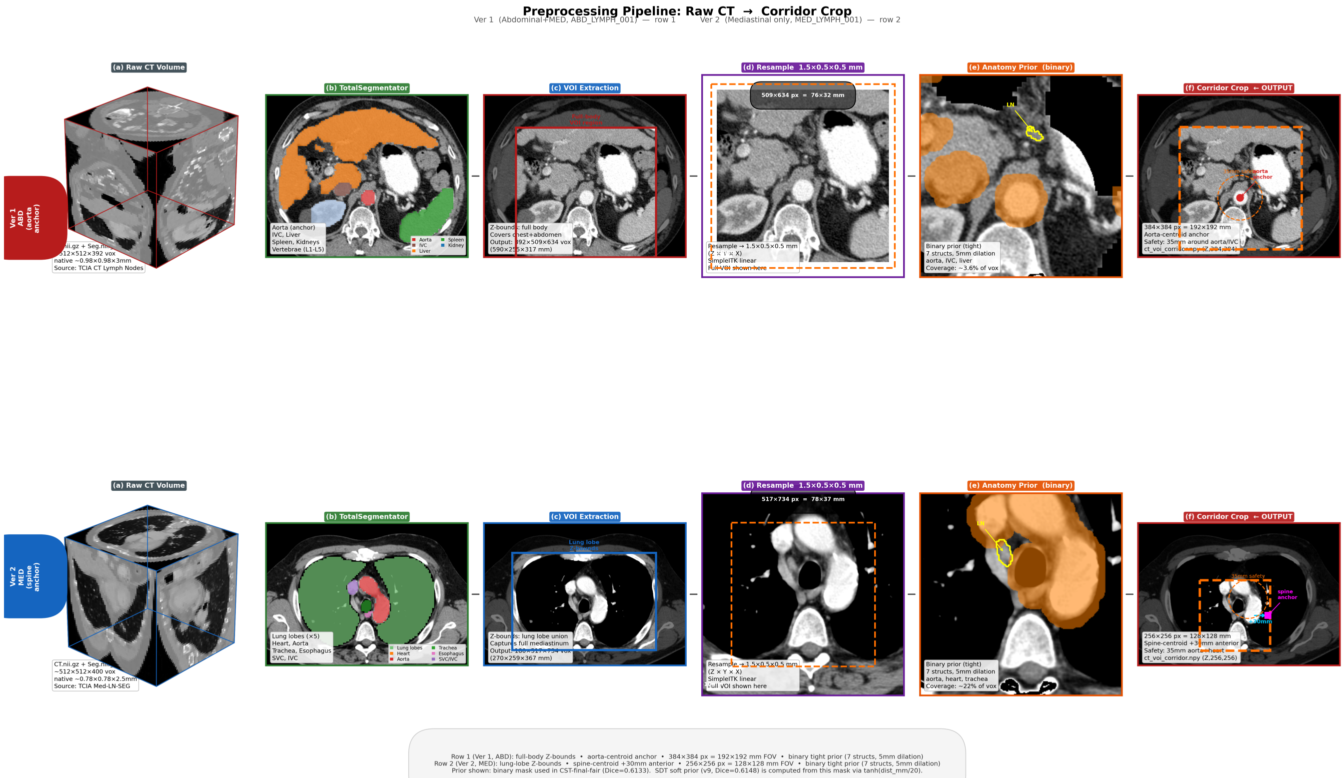

FIGURE 1. Preprocessing pipeline: (a) raw CT, (b) TotalSegmentator organ segmentation, (c) VOI bounding box, (d) resampled VOI at 1.5 × 0.5 × 0.5 mm, (e) anatomy prior M_{prior} with GT contour (yellow), (f) final corridor crop. Row 1: Ver1 abdominal patient. Row 2: Ver2 mediastinal patient.

the *full corridor volume* over the soft-tissue HU window $[-50, 300]$ HU:

$$I_{\text{norm}}(p) = \text{clip}\left(\frac{I(p) - \mu_\alpha}{\sigma_\alpha + \varepsilon},\ -3,\ 3\right), \tag{2}$$

where $\varepsilon = 10^{-6}$. The clipped output is mapped linearly to $[0, 1]$. Statistics are computed per patient and remain fixed between training and inference, with no dependence on batch composition. All models are trained from scratch on CT data.

C. 2.5D INPUT CONSTRUCTION

Fig. 2 illustrates the 6-channel CST input tensor assembled from the corridor volumes.

1) Context Slice Assembly

For each target slice z^*, five CT context slices are extracted as input channels:

$$\mathbf{C}(z^*) = \left[I_{z^*-2},\, I_{z^*-1},\, I_{z^*},\, I_{z^*+1},\, I_{z^*+2}\right], \tag{3}$$

capturing a ± 3.0 mm axial range at 1.5 mm slice spacing. Out-of-bound indices are clamped to the volume extent. The five-slice window was selected via ablation (Section IV-D); wider contexts provide diminishing returns because adjacent slices at 1.5 mm spacing are highly correlated.

2) Anatomy Prior Channel and Final Tensor

The 6th channel (appended to the five CT context channels) carries the binary anatomy prior M_{prior} from (1), providing a spatial map of the lymph node neighbourhood. The final input tensor is:

$$\mathbf{X}(z^*) = \left[\mathbf{C}(z^*),\ M_{\text{prior}}(\cdot, z^*)\right] \in \mathbb{R}^{6\times256\times256}. \tag{4}$$

3) Training Sampling

Positive slices are oversampled with weight 5.0 relative to negative slices to counteract the severe class imbalance (median foreground fraction $\approx$0.35%). This weight was selected on fold 0 validation Dice over $\{2, 5, 10\}$; weight 5.0 yielded $+0.008$ Dice over weight 2.0 and $+0.005$ over weight 10.0. The positive-slice list is computed directly from the corridor segmentation mask to avoid ghost-positive artefacts.

D. MODEL ARCHITECTURES

We propose the Cross-Slice Transformer (CST) for adaptive inter-slice modelling in 2.5D CT analysis and compare it against three baseline segmentation architectures: standard CNN (U-Net), hybrid CNN–transformer (TransUNet), and pure transformer (SwinUNet). All 2.5D models process the 256×256 corridor input directly without resizing. All 2.5D models receive a 6-channel input comprising five consecutive CT slices at offsets $\{-2, -1, 0, +1, +2\}$ plus one binary anatomy prior channel, and are trained from scratch without pretrained weights.

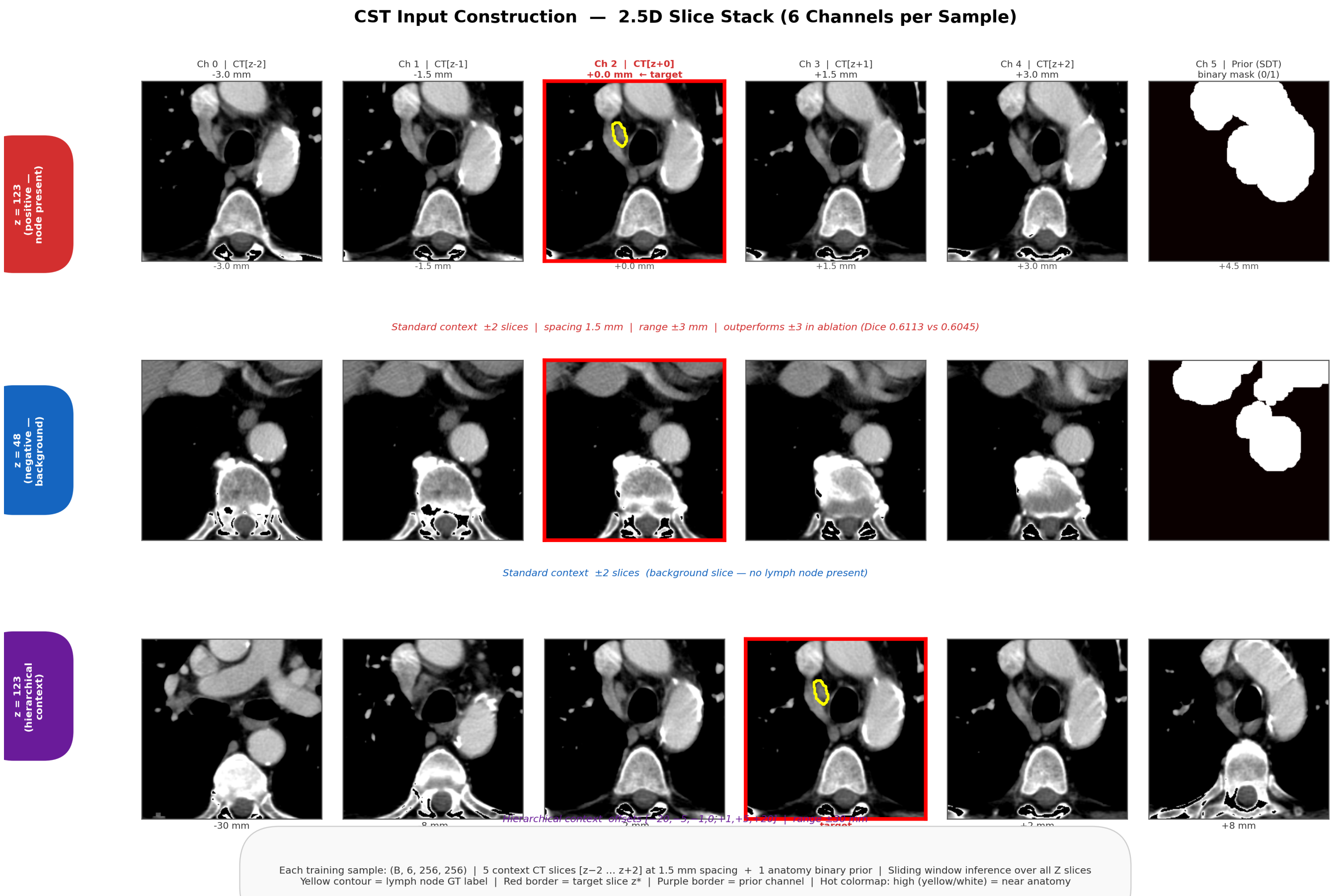

FIGURE 2. **CST 2.5D input construction (6 channels per sample). Row 1: positive slice ($z^* = 123$, yellow contour = GT, red border = target). Row 2: negative slice ($z^* = 48$, no node present). Row 3: wider-range context used in ablation only. Ch 5 (purple border): binary anatomy prior M_{prior}.**

1) U-Net

U-Net [7] with ResNet-50 [8] encoder provides the CNN baseline, implemented via the segmentation_models_pytorch package [26]. The encoder extracts hierarchical representations through five stages with channel progression [64, 256, 512, 1024, 2048]. The decoder reconstructs spatial detail through four upsampling stages using bilinear interpolation followed by convolutional layers, with skip connections preserving fine-grained spatial information from the encoder.

2) TransUNet

TransUNet [9] combines CNN feature extraction with transformer-based global context modelling. We implement a hybrid architecture using a ResNet-50 CNN encoder from `torchvision` and a Vision Transformer (ViT-Base-Patch16) from `timm` [27]. The ViT backbone employs 12 transformer layers with 12 attention heads, 768-dimensional embeddings, and patch size 16. The CNN decoder uses channel progression [256, 128, 64, 16] for spatial upsampling. The model is instantiated with `img_size` = 256, so the full 256×256 corridor input is processed without resizing.

3) SwinUNet

SwinUNet [10] uses a hierarchical Swin Transformer as the encoder. We implement a hybrid architecture combining a SwinV2-Tiny encoder from `segmentation_models_pytorch` with a standard U-Net CNN decoder. The encoder uses window size 8, patch size 4, and [2, 2, 6, 2] transformer blocks per stage with [3, 6, 12, 24] attention heads and 96-dimensional embeddings. The decoder uses channel progression [256, 128, 64, 32, 16]. The model is configured for 256×256 input and processes the corridor volume without resizing.

4) Cross-Slice Transformer (CST)

The Cross-Slice Transformer learns adaptive slice importance through statistical descriptor extraction and transformer-based attention for explicit inter-slice modelling. The architecture integrates three components with a lightweight U-Net decoder, as illustrated in Fig. 3.

Statistical Descriptor Extraction. For each input of six channels $\mathbf{X} \in \mathbb{R}^{B \times 6 \times H \times W}$ (five CT slices and one anatomy prior), we extract compact descriptors encoding channel-level intensity distribution and inter-channel coherence. For each

TABLE 1. Dataset statistics for Ver1 (multi-region, 296 fully annotated patients) and Ver2 (mediastinal, 206 fully annotated patients). Node counts reflect all fully annotated patients from both source collections. Class imbalance uses median (IQR), excluding four patients with empty corridor masks. Size-stratum percentages are rounded to the nearest integer; exact node counts per stratum are given in Table 4.

Statistic	Ver1 (Multi-region)	Ver2 (Mediastinal)
Patients (fully annotated)	296	206
Slices per volume	243 ± 99	182 ± 20
Lymph node instances	1,770	1,250
LN per patient	6.0	6.1 ± 4.5
Short-axis diameter (mm)	14.9 ± 10.7	16.9 ± 10.5
Small / Medium / Large (%)	35 / 22 / 43	43 / 18 / 39
Positive slice ratio	0.254 ± 0.166	0.300 ± 0.166
FG voxel fraction (median, %)	0.24	0.35
BG:FG ratio (median, IQR)	414 : 1 [186, 1,243]	286 : 1 [135, 582]
FG mean HU	46 ± 21	43 ± 22
BG mean HU	-243 ± 66	-224 ± 62

channel $s \in \{1, \dots, 6\}$:

$$\mu_s = \frac{1}{HW} \sum_{h,w} X_{s,h,w}, \quad \sigma_s = \sqrt{\frac{1}{HW} \sum_{h,w} (X_{s,h,w} - \mu_s)^2}, \tag{5}$$

$$\delta_s = \begin{cases} \frac{1}{HW} \sum_{h,w} |X_{2,h,w} - X_{1,h,w}| & s = 1 \\ \frac{1}{HW} \sum_{h,w} |X_{s,h,w} - X_{s-1,h,w}| & s > 1 \end{cases} \tag{6}$$

The descriptor vector $\mathbf{d}_s = [\mu_s, \sigma_s, \delta_s, \delta_s/(\mu_s + \varepsilon)] \in \mathbb{R}^4$ encodes both local statistics and inter-channel variation.

Transformer-Based Attention. Descriptor vectors are projected into a 128-dimensional embedding space and processed through a transformer encoder:

$$\mathbf{T}_s = \mathrm{GELU}(\mathrm{LN}(\mathbf{W}_e \, \mathbf{d}_s)), \quad \mathbf{W}_e \in \mathbb{R}^{128\times 4}, \tag{7}$$

$$\mathbf{T}' = \mathrm{TransformerEncoder}([\mathbf{T}_1, \dots, \mathbf{T}_6]), \tag{8}$$

$$w_s = \frac{\exp(\mathbf{W}_w \, \mathbf{T}'_s)}{\sum_{j=1}^{6} \exp(\mathbf{W}_w \, \mathbf{T}'_j)}, \quad \mathbf{W}_w \in \mathbb{R}^{1\times 128}, \tag{9}$$

where w_s is the learned importance weight for channel s. The transformer encoder uses 3 layers, 8 attention heads, hidden dimension 128, feedforward dimension 256, and dropout 0.1.

Contextual Fusion. The learned weights produce an adaptively fused representation combined with spatial difference maps:

$$\mathbf{F}_{\text{fused}} = \sum_{s=1}^{6} w_s \cdot \mathbf{X}_s, \tag{10}$$

$$\mathbf{D}_{\text{spatial}} = \frac{1}{5} \sum_{s=2}^{6} |\mathbf{X}_s - \mathbf{X}_{s-1}|, \tag{11}$$

$$\mathbf{C} = [\mathbf{X}_{1:6}, \, \mathbf{F}_{\text{fused}}, \, \mathbf{D}_{\text{spatial}}] \in \mathbb{R}^{B\times 8\times H\times W}. \tag{12}$$

The context tensor $\mathbf{C}$ is processed through convolutional layers and fed to a U-Net decoder with base channels 64 and progression [64, 128, 256, 512] across four downsampling levels, yielding a binary segmentation mask for the centre slice.

Interpretability Signal. The model additionally computes the normalised inter-channel difference energy:

$$e_s = \frac{\delta_s}{\sum_{j=1}^{6} \delta_j + \varepsilon}, \tag{13}$$

which serves as an interpretability signal — a channel with high e_s exhibits greater content change relative to its neighbour and would intuitively deserve higher attention weight w_s. During development, a consistency regularisation term $\mathcal{L}_{\text{cons}} = \frac{1}{6} \sum_{s=1}^{6} (w_s - e_s)^2$ was evaluated as an auxiliary loss to explicitly align w_s with e_s; however, it produced no improvement in validation Dice and was excluded from the final training objective. The model outputs both w_s and e_s and their agreement can be inspected at inference to assess whether the transformer attends to the most informative channels.

Table **??** summarises the computational complexity of all evaluated architectures. CST (7.97 M parameters) is more compact than TransUNet (99.21 M) and U-Net (32.54 M), while its higher FLOPs (41.86 G) reflect the statistical descriptor computation and transformer attention over all six input channels.

E. TRAINING CONFIGURATION

All 2.5D models were trained with identical hyperparameters to ensure that observed performance differences reflect architectural design rather than optimisation choices. nnUNet-3D uses its default self-configured training protocol [12].

1) Optimisation

All 2.5D models were trained using AdamW [28] with learning rate 10^{-4}, cosine decay to 10^{-6}, 10-epoch linear warmup, weight decay 10^{-4}, and gradient clipping at 1.0. Batch size was 16. Training ran for up to 300 epochs with early stopping (patience 20 epochs) monitored on validation Dice. Exponential moving average (EMA) of weights with decay 0.999 was maintained for all 2.5D models and used at inference. All models were trained from scratch without pretrained weights.

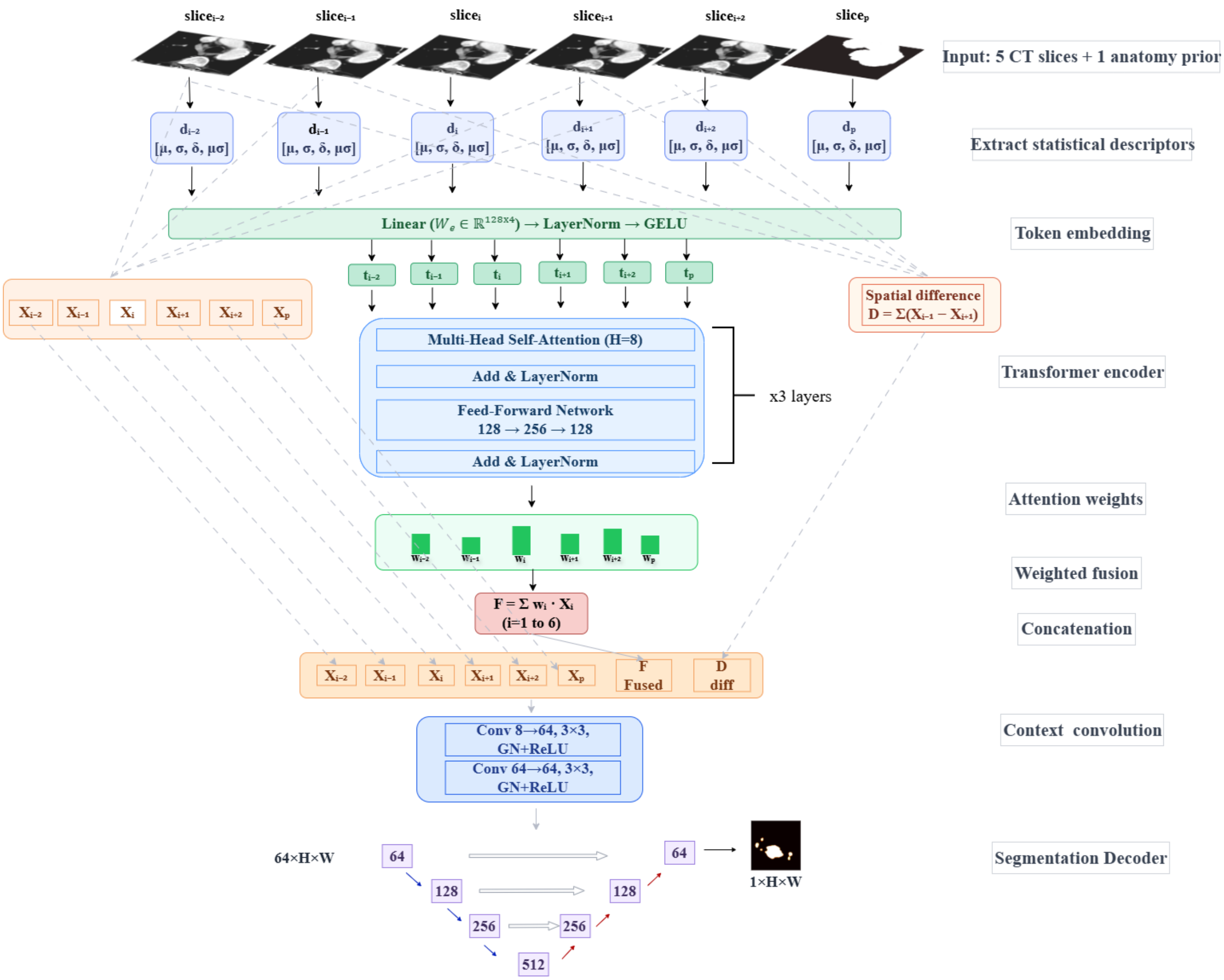

FIGURE 3. Cross-Slice Transformer (CST) architecture. Statistical descriptors are extracted from all six input channels ($s = 1$ to 6), importance weights computed via transformer attention, and channels adaptively fused before U-Net decoding.

2) Loss Function

All 2.5D models use the composite loss:

$$\mathcal{L} = 0.40 \cdot \mathcal{L}_{\text{Tversky}} + 0.30 \cdot \mathcal{L}_{\text{SoftDice}} + 0.20 \cdot \mathcal{L}_{\text{Boundary}} + 0.10 \cdot \mathcal{L}_{\text{Focal}}, \tag{14}$$

where the Tversky term uses $\alpha = 0.30$, $\beta = 0.70$:

$$\mathcal{L}_{\text{Tversky}} = 1 - \frac{TP + 1}{TP + 0.30 \cdot FP + 0.70 \cdot FN + 1}, \tag{15}$$

penalising false negatives 2.3× more than false positives to prioritise recall, which is clinically motivated by the higher cost of a missed lymph node. The SoftDice term provides global volume-overlap regularisation. The boundary loss uses the signed distance transform of the ground truth to penalise predictions far from the GT boundary, improving delineation of small nodes with low-contrast borders. The Focal term ($\gamma = 1.0$, $\alpha = 0.75$) down-weights easy negatives. This composite loss is applied identically to all 2.5D models including CST.

TABLE 2. Data augmentation applied to all 2.5D models during training.

Augmentation	Probability	Parameters
Horizontal flip	0.5	—
Rotation	0.3	±10°
Scale	0.5	±5%
Gaussian blur	0.3	—
Gamma correction	0.3	[0.8, 1.2]
HU intensity shift	0.5	±15 HU
Gaussian noise	0.3	$\sigma = 5$ HU
Copy-paste	0.5	—

3) Data Augmentation

Augmentations were applied online identically across all 2.5D models (Table 2). Copy-paste augmentation [29] increases the frequency of rare positive examples by pasting lymph node crops from other training patients, directly addressing foreground sparsity. Foreground slices were additionally oversampled at 5× the rate of background slices during batch construction.

F. EVALUATION METRICS

1) Segmentation Metrics

The Dice Similarity Coefficient (DSC) is the primary metric:

$$\mathrm{DSC} = \frac{2\,TP}{2\,TP + FP + FN}. \tag{16}$$

Dice is computed per patient over the full 3D corridor volume, then averaged across patients (each patient weighted equally). Edge cases: GT empty and pred empty $\rightarrow$ DSC = 1; one side empty $\rightarrow$ DSC = 0. Complementary metrics (IoU, Precision, Recall) follow the same patient-level aggregation. No test-time augmentation or postprocessing is applied; EMA weights are used at inference for all 2.5D models.

2) Size-Stratified Analysis

Performance is stratified by lymph node short-axis diameter (RECIST 1.1 standard [5]):

$$d_{\mathrm{short}} = \min(\mathrm{bbox}_x,\, \mathrm{bbox}_y) \times 0.5\,\mathrm{mm/px}, \tag{17}$$

using the in-plane bounding box only (Z extent excluded). Strata: Small < 10 mm, Medium 10–15 mm, Large ≥ 15 mm. Per-node Dice is computed within each node's 3D bounding box to avoid cross-node FP contamination in multi-node patients.

3) Node Detection

A GT connected component is considered detected if its IoU with any predicted component, computed within the node's 3D bounding box, satisfies IoU ≥ 0.1. The threshold 0.1 follows the convention for small-structure detection (LUNA16, LIDC-IDRI) and is appropriate given the sub-centimetre size of many lymph nodes.

IV. RESULTS

A. OVERALL SEGMENTATION PERFORMANCE

Table 3 presents fold-level segmentation performance on Ver1 and Ver2. On the mediastinal-focused Ver2, CST-2.5D achieves the highest Dice among 2.5D models (0.5428 ± 0.0440), while nnUNet-3D achieves higher overall Dice (0.5891 ± 0.0456) as expected given its full volumetric context advantage. Among 2.5D architectures, CST leads TransUNet by 6.4 percentage points, UNet by 2.2 percentage points, and SwinUNet by 1.1 percentage points on Ver2 Dice. On the multi-region Ver1, nnUNet-3D leads (0.5146 ± 0.0323) and CST ranks first among 2.5D models (0.4963 ± 0.0292). Fig. 4 summarises the fold-level Dice for all models on both datasets, with error bars showing inter-fold standard deviation. All models improve when moving from Ver1 to Ver2, confirming that anatomically focused single-region training yields consistently higher performance regardless of architecture. The improvement ranges from $+0.047$ (CST) to $+0.152$ (TransUNet) across 2.5D models, and $+0.075$ for nnUNet-3D. TransUNet shows the largest gain ($+0.152$) while CST shows the smallest ($+0.047$), suggesting that CST performance is more consistent across dataset configurations rather than being particularly sensitive to anatomical scope.

TABLE 3. Overall segmentation performance (mean ± std across 5 folds, fold-level). Each value is the mean of per-fold patient-level means; std reflects inter-fold variance confirming training stability. Best value per column and dataset in bold.

Dataset	Model	Dice	Precision	Recall
Ver1	nnUNet-3D	**0.5146 ± 0.0323**	**0.5203 ± 0.0311**	**0.6205 ± 0.0409**
	UNet-2.5D	0.4134 ± 0.0104	0.4224 ± 0.0294	0.4639 ± 0.0341
	TransUNet-2.5D	0.3266 ± 0.0493	0.3517 ± 0.0652	0.3822 ± 0.0958
	SwinUNet-2.5D	0.4381 ± 0.0251	0.4259 ± 0.0454	0.5080 ± 0.0417
	CST-2.5D (ours)	0.4963 ± 0.0292	0.4678 ± 0.0447	0.5848 ± 0.0394
Ver2	nnUNet-3D	**0.5891 ± 0.0456**	**0.6648 ± 0.0239**	0.6032 ± 0.0549
	UNet-2.5D	0.5213 ± 0.0231	0.5299 ± 0.0212	0.5731 ± 0.0507
	TransUNet-2.5D	0.4784 ± 0.0385	0.4913 ± 0.0798	0.5429 ± 0.0764
	SwinUNet-2.5D	0.5316 ± 0.0296	0.5315 ± 0.0337	0.6039 ± 0.0243
	CST-2.5D (ours)	0.5428 ± 0.0440	0.5368 ± 0.0658	**0.6179 ± 0.0243**

On Ver2, CST achieves the highest recall (0.6179 ± 0.0243) among all models, indicating stronger sensitivity at the cost of lower precision (0.5368 ± 0.0658) compared to nnUNet (0.6648 ± 0.0239). On Ver1, nnUNet leads in recall (0.6205 ± 0.0409) while 2.5D models show a consistent pattern of higher recall than precision, reflecting the effect of the recall-biased Tversky loss ($\beta = 0.70$).

B. SIZE-STRATIFIED ANALYSIS

Table 4 presents performance stratified by lymph node short-axis diameter. Across all models and both datasets, performance increases monotonically with node size, confirming that small nodes remain the primary challenge in automated lymph node segmentation.

1) Small Nodes ($<$10 mm)

Small nodes represent the most challenging category for all architectures. On Ver2, CST achieves the highest node Dice (0.345 ± 0.054) and detection rate (288/537, 53.6%), outperforming nnUNet-3D (231/537, 43.0%) and all 2.5D baselines. On Ver1, CST leads among 2.5D models (291/612, 47.5%) while nnUNet-3D achieves 238/612 (38.9%) detection. TransUNet performs worst in this category on both datasets (Ver2: 27.6%, Ver1: 24.3%), reflecting limited capacity to localise sub-centimetre structures with patch-based global context modelling.

2) Medium Nodes (10–15 mm)

Detection rates improve substantially in this range. On Ver2, CST leads detection (184/222, 82.9%) followed by SwinUNet (168/222, 75.7%) and nnUNet-3D (160/222, 72.1%). On Ver1, CST again leads among 2.5D models (282/381, 74.0%) while nnUNet-3D achieves 273/381 (71.7%). Node Dice converges across architectures in this range (0.34–0.52 on Ver2), indicating that medium nodes are sufficiently large for most models to localise reliably.

3) Large Nodes ($\geq$15 mm)

All models achieve their best performance on large nodes. On Ver2, CST leads with node Dice 0.656 ± 0.020 and detection 454/491 (92.5%), followed by SwinUNet (439/491, 89.4%) and nnUNet-3D (425/491, 86.6%). On Ver1, nnUNet-3D leads (665/777, 85.6%) while CST achieves 656/777 (84.4%).

The convergence of performance across architectures for large nodes confirms that architectural differences are most consequential for small and medium nodes. Figs. 5 and 6 show representative successful and failure cases, illustrating the precision-recall trade-off discussed above.

C. CLINICAL IMPLICATIONS

The results reveal two distinct performance tiers. nnUNet-3D consistently leads on overall Dice across both datasets, reflecting the advantage of full volumetric context. Among 2.5D models, CST leads on Ver2 and ranks first on all size strata on Ver2 and on small and medium strata on Ver1, demonstrating that explicit cross-slice attention provides consistent benefit for node localisation regardless of size. The apparent discrepancy between overall Dice rankings and per-node size-stratified rankings warrants explanation. Overall patient-level Dice pools all voxels globally: a model with higher voxel-level recall will accumulate more true positives across the whole volume, pushing overall Dice upward even if it also generates more false positives. Per-node Dice, by contrast, is computed within each node's 3D bounding box independently. As a full 3D model, nnUNet predictions can bleed slightly beyond exact node boundaries, increasing false positives within each bounding box and reducing per-node Dice even when the node is correctly detected. CST, guided by the anatomy prior channel, tends to produce spatially tighter predictions concentrated near the node boundary, resulting in fewer false positives within each bounding box and higher per-node Dice despite lower overall voxel-level recall. This explains why CST leads on per-node detection and per-node Dice across all size strata while nnUNet-3D leads on overall patient-level Dice. The inter-fold standard deviations across all models (0.010–0.066) reflect training stability, while the high per-patient variance visible in the qualitative figures indicates case-dependent difficulty driven by sub-centimetre or isolated nodes. Notably, CST achieves the highest recall on Ver2 (0.6179) at the cost of lower precision (0.5368), a trade-off that is clinically preferable in a staging context where missed nodes carry greater consequence than false positives [5]. Architecture selection should be guided by clinical task and available resources: nnUNet-3D for maximum Dice when GPU memory permits full 3D inference; CST-2.5D for memory-efficient deployment with strong recall on focused anatomical regions; and SwinUNet-2.5D as the strongest alternative when CST is not used, particularly on multi-region data.

Qualitative analysis of failure cases (Fig. 6) reveals four recurring patterns across all models. First, vascular false positives: nodes adjacent to vessels with similar HU values generate large false positive regions; only nnUNet-3D and CST produce any true positive overlap in these cases, reflecting the benefit of 3D context and anatomy-guided spatial priors respectively. Second, sub-centimetre scatter: nodes below 5 mm are near the resolution limit at 1.5 mm axial spacing; nnUNet partially localises these while 2.5D models frequently score zero. Third, CST over-segmentation: in some cases CST merges adjacent nodes into a single connected blob, sacrificing node individuality for higher recall — a direct consequence of the recall-biased Tversky loss. Fourth, anatomy prior boundary failures: nodes located at the edge of or outside the corridor produce complete failure across all models, indicating that corridor width and prior coverage are the binding constraint for a small fraction of lateral nodes.

D. ABLATION STUDY

All ablation experiments are conducted on Ver2 (mediastinal, fold 0) using the same training protocol as the main experiments; fold 0 was selected to reduce computational cost, as running all five folds per ablation variant would require 60 additional training runs. Dice is reported on the fold 0 validation set (41 patients). The full CST fold 0 Dice (0.6133, Table 6) is higher than the 5-fold pooled mean in Table 3 (0.5428 ± 0.0440) due to natural inter-fold variance in this small cohort; the two figures measure different patient subsets and should not be compared directly.

1) Context Window Size

Table 5 evaluates the effect of the number of input CT slices on segmentation performance. A single-slice input (no inter-slice context) yields Dice = 0.5829, confirming that neighbouring slices carry useful information for lymph node localisation. Performance improves monotonically from 1 to 5 slices, then drops slightly at 7 slices (0.6045 vs 0.6113), indicating diminishing returns from wider contexts. This is expected: at 1.5 mm axial spacing, slices beyond ± 2 offsets (± 3 mm) are highly correlated with the centre slice and add little discriminative signal while increasing input dimensionality. We therefore adopt a 5-slice context (± 2 offsets) as our final configuration.

2) CST Component Ablation

Table 6 isolates the contribution of each architectural component of CST by ablating one element at a time. Removing the interaction descriptor (the normalised ratio $\delta_s/(\mu_s + \varepsilon)$, reducing each descriptor from 4 to 3 elements) degrades Dice by 0.0081, demonstrating that explicit inter-slice contrast encoding improves the transformer's ability to discriminate informative from redundant slices. Replacing the transformer with uniform fixed weights ($1/5$ per slice, no learned attention) degrades Dice by 0.0060. The model still benefits from multi-slice fusion but loses the ability to dynamically suppress uninformative slices, confirming that adaptive weighting is the primary mechanism of the transformer component. Replacing the transformer with a two-layer MLP produces the largest degradation (-0.0397), reducing Dice to 0.5736. This result highlights that global inter-slice reasoning through self-attention is critical: an MLP processes each slice's descriptor independently and cannot model dependencies across the full context window, leading to substantially worse slice selection.

TABLE 4. Size-stratified segmentation performance (mean ± std across 5 folds, fold-level). Dice computed over the full 3D patient volume for voxels belonging to each size stratum. Detection: IoU ≥ 0.1 within node 3D bounding box. Size by RECIST short-axis: $\min(\text{bbox}_x, \text{bbox}_y) \times 0.5$ mm/px. Best Dice value per column and stratum in bold.

Model	Stratum	Ver1		Ver2	
		Dice	Detection	Dice	Detection
nnUNet-3D	<10 mm	0.265 ± 0.034	238/612 (38.9%)	0.301 ± 0.036	231/537 (43.0%)
	10–15 mm	$\mathbf{0.523 \pm 0.036}$	273/381 (71.7%)	0.507 ± 0.066	160/222 (72.1%)
	≥ 15 mm	$\mathbf{0.632 \pm 0.031}$	**665/777 (85.6%)**	0.639 ± 0.057	425/491 (86.6%)
UNet-2.5D	<10 mm	0.184 ± 0.056	183/612 (29.9%)	0.234 ± 0.067	209/537 (38.9%)
	10–15 mm	0.310 ± 0.054	189/381 (49.6%)	0.395 ± 0.047	151/222 (68.0%)
	≥ 15 mm	0.457 ± 0.047	534/777 (68.7%)	0.569 ± 0.038	420/491 (85.5%)
TransUNet-2.5D	<10 mm	0.144 ± 0.045	149/612 (24.3%)	0.151 ± 0.031	148/537 (27.6%)
	10–15 mm	0.249 ± 0.086	159/381 (41.7%)	0.341 ± 0.077	140/222 (63.1%)
	≥ 15 mm	0.370 ± 0.075	464/777 (59.7%)	0.524 ± 0.056	414/491 (84.3%)
SwinUNet-2.5D	<10 mm	0.238 ± 0.044	233/612 (38.1%)	0.273 ± 0.043	230/537 (42.8%)
	10–15 mm	0.359 ± 0.078	215/381 (56.4%)	0.460 ± 0.037	168/222 (75.7%)
	≥ 15 mm	0.498 ± 0.042	571/777 (73.5%)	0.602 ± 0.046	439/491 (89.4%)
CST-2.5D (ours)	<10 mm	$\mathbf{0.312 \pm 0.031}$	**291/612 (47.5%)**	$\mathbf{0.345 \pm 0.054}$	**288/537 (53.6%)**
	10–15 mm	0.476 ± 0.020	**282/381 (74.0%)**	$\mathbf{0.525 \pm 0.027}$	**184/222 (82.9%)**
	≥ 15 mm	0.582 ± 0.036	656/777 (84.4%)	$\mathbf{0.656 \pm 0.020}$	**454/491 (92.5%)**

Dice: mean ± std across 5 folds (fold-level). Detection rates are pooled across all folds.

TABLE 5. Effect of context window size on fold 0 validation Dice (Ver2). Best result in bold.

CT Slices	Offsets	Dice
1	$\{0\}$	0.5829
3	$\{-1, 0, +1\}$	0.5950
5 (ours)	$\{-2, -1, 0, +1, +2\}$	**0.6113**
7	$\{-3, \ldots, +3\}$	0.6045

TABLE 6. CST component ablation on fold 0 validation Dice (Ver2). ´ is relative to the full model.

Variant	Dice	Δ
Full CST (ours)	**0.6133**	—
w/o Interaction term	0.6052	−0.0081
Fixed weights (1/5)	0.6073	−0.0060
MLP fusion	0.5736	−0.0397

3) Intensity Normalisation Strategy

Table 7 compares three normalisation strategies. Per-case Z-score normalisation over the soft-tissue HU window $[-50, 300]$ HU achieves the highest Dice (0.6133). Dataset-wide normalisation performs slightly worse (−0.0016), as it cannot account for scanner-specific HU shifts that vary substantially across the multi-institution TCIA data. Fixed HU windowing $[-160, 240]$ HU shows the largest degradation (−0.0075), confirming that rigid intensity mapping is poorly suited to heterogeneous CT protocols where soft-tissue HU distributions differ across acquisition sites.

TABLE 7. Effect of intensity normalisation strategy on fold 0 validation Dice (Ver2). ´ relative to per-case Z-score.

Strategy	Dice	Δ
Per-case Z-score (ours)	**0.6133**	—
Dataset-wide mean/std	0.6117	−0.0016
Fixed HU window $[-160, 240]$	0.6058	−0.0075

V. DISCUSSION

A. KEY FINDINGS

Among 2.5D models, CST achieves the highest Dice on Ver2 (0.5428 ± 0.0440) and the highest recall across both datasets, demonstrating that explicit cross-slice attention effectively leverages inter-slice context. nnUNet-3D achieves higher overall Dice on both datasets, which is consistent with its full volumetric context advantage and does not diminish the 2.5D contribution. The meaningful architectural comparison is therefore among the four 2.5D models, where CST ranks first on both datasets. On Ver2, CST leads SwinUNet by 1.1 percentage points on Dice (0.5428 vs 0.5316), with additional advantages in detection rate (53.6% vs 42.8% for small nodes) and parameter efficiency (7.97 M vs 25.22 M parameters). The size-stratified analysis provides a clearer picture of CST's contribution. CST leads all models including nnUNet-3D on node detection rate across every size stratum on Ver2: 53.6% for small nodes, 82.9% for medium, and 92.5% for large. This detection advantage reflects CST's higher recall (0.6179 vs 0.6032 for nnUNet-3D), which comes at the cost of lower precision (0.5368 vs 0.6648). The higher recall means CST pro-

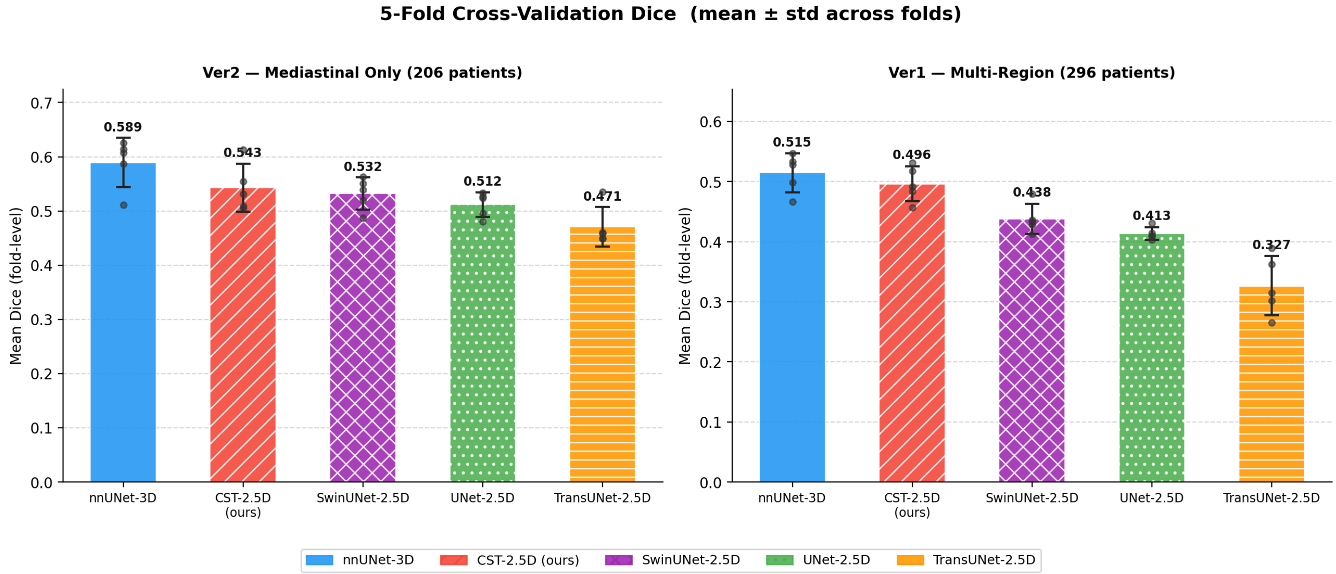

FIGURE 4. Five-fold cross-validation Dice for Ver2 (left) and Ver1 (right). Error bars: inter-fold std. Scatter points: individual fold means.

duces more predictions that overlap GT nodes at IoU ≥ 0.1, which drives the detection rate advantage. In a clinical staging context where a missed node may represent undetected metastasis, higher recall is the more clinically relevant property. To assess the statistical reliability of observed Dice differences, paired t-tests were conducted at the patient level, comparing per-patient Dice scores between CST and each baseline on identical held-out test sets ($n = 206$ for Ver2, $n = 296$ for Ver1). 95% confidence intervals (normal approximation) for the mean pairwise difference are reported alongside each p-value. On Ver2, nnUNet-3D is statistically significantly better than CST ($\Delta = -0.063$, 95 % CI $[-0.084, -0.042]$, $p < 0.001$), confirming that the volumetric context advantage of full 3D processing produces a genuine and consistent improvement over 2.5D approaches. Among 2.5D models, CST significantly outperforms UNet ($\Delta = +0.025$, 95 % CI $[+0.009, +0.041]$, $p = 0.003$) and TransUNet ($\Delta = +0.067$, 95 % CI $[+0.051, +0.083]$, $p < 0.001$). The difference between CST and SwinUNet on Ver2 is not statistically significant ($\Delta = +0.001$, 95 % CI $[-0.014, +0.017]$, $p = 0.871$), confirming that their aggregate Dice scores are statistically equivalent on this dataset. CST's practical advantage over SwinUNet on Ver2 is therefore better characterised by node-level detection rate (53.6% vs 42.8% for small nodes) and parameter efficiency (7.97 M vs 25.22 M) than by aggregate Dice. On Ver1, nnUNet-3D again significantly outperforms CST ($\Delta = -0.060$, 95 % CI $[-0.077, -0.044]$, $p < 0.001$). Among 2.5D models, CST significantly outperforms all baselines: SwinUNet ($\Delta = +0.046$, 95 % CI $[+0.033, +0.059]$, $p < 0.001$), UNet ($\Delta = +0.065$, 95 % CI $[+0.050, +0.081]$, $p < 0.001$), and TransUNet ($\Delta = +0.153$, 95 % CI $[+0.134, +0.171]$, $p < 0.001$). The consistently significant margins on Ver1 confirm that CST's advantage among 2.5D models is not fold-dependent but holds across the full patient population. The patient-level Δ values reflect median pairwise differences per patient and therefore differ slightly from the fold-mean differences reported in Table 3; both estimators are consistent in direction and magnitude, with patient-level differences tending to be larger because fold-averaging compresses variance in heterogeneous cohorts.

B. IMPACT OF DATASET ANATOMICAL CONFIGURATION

All models improve when moving from multi-region Ver1 to mediastinal-focused Ver2, with Dice gains ranging from $+0.047$ (CST) to $+0.152$ (TransUNet) among 2.5D models, and $+0.075$ for nnUNet-3D. This consistent pattern across all five architectures confirms that the performance difference is driven by dataset composition rather than any single model property. TransUNet benefits most ($+0.152$) while CST shows the smallest gain ($+0.047$), suggesting that CST generalises more consistently across dataset configurations and is less dependent on anatomical homogeneity. We note that Ver2 is a strict subset of Ver1, so the comparison conflates anatomical scope with dataset size. A fully controlled experiment training on a 206-patient subset of Ver1 would be required to isolate the effect of anatomical homogeneity from dataset size, which we identify as a limitation and a direction for future work.

C. LIMITATIONS

Several limitations should be noted when interpreting these results. Both datasets originate from the TCIA repository and together comprise 296 patients for Ver1 and 206 patients for Ver2, which, while sufficient for cross-validation evaluation, may limit generalisation to unseen scanner protocols and patient populations; larger multi-institutional cohorts are needed to confirm the observed performance trends. Since Ver2 is a strict subset of Ver1, the two benchmarks are not indepen-

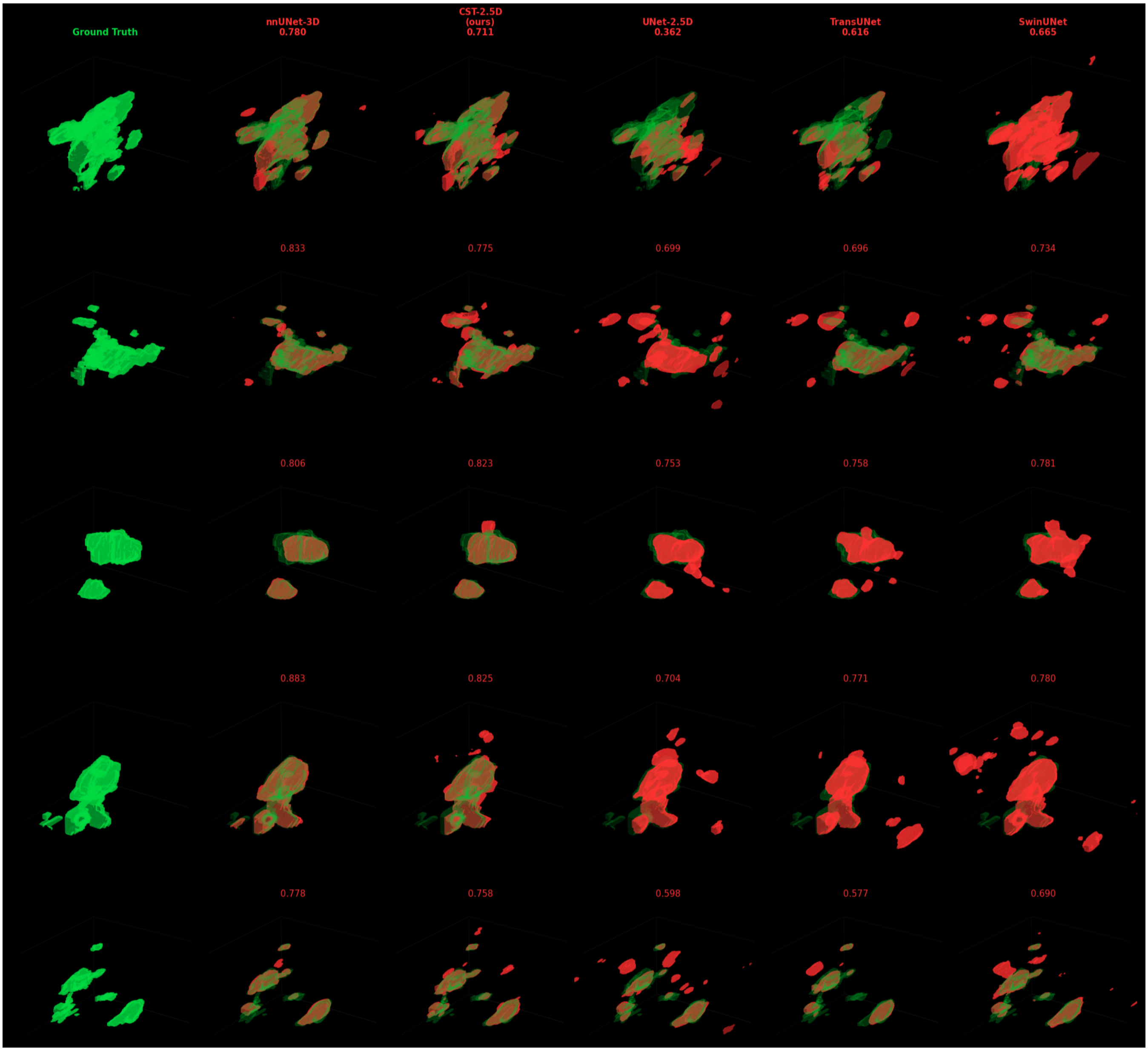

FIGURE 5. Representative successful segmentations on five Ver2 patients. GT: green; prediction: red; overlap: orange. Per-patient Dice shown above each column. Columns: Ground Truth, nnUNet-3D, CST-2.5D (ours), UNet-2.5D, TransUNet-2.5D, SwinUNet-2.5D.

dent cohorts, and the Ver1-to-Ver2 improvement conflates anatomical scope with dataset size. Nodes under 10 mm are below the RECIST 1.1 suspicious size threshold and are not routinely assessed in current clinical practice; performance on this stratum reflects a research objective motivated by earlier staging potential rather than an immediate clinical requirement. Finally, evaluation is conducted against reference annotations only, without inter-observer agreement analysis or radiologist reader studies, and multi-centre external validation is required before clinical deployment.

The 2.5D baseline models were trained from scratch without pretrained weights to ensure a controlled comparison with CST, which lacks a natural large-scale pretraining equivalent; standard TransUNet and SwinUNet implementations typically benefit from ImageNet initialisation, and training from scratch on a dataset of this size may place these architectures below their reported peak performance. The observed performance gaps should therefore be interpreted as reflecting both architectural differences and optimisation constraints, and future work should evaluate pretrained variants under a unified fine-tuning protocol.

D. CLINICAL TRANSLATION

CST's higher recall (0.6179 on Ver2) makes it preferable to other 2.5D models in staging workflows where sensitivity is prioritised over precision. Its compact parameter count

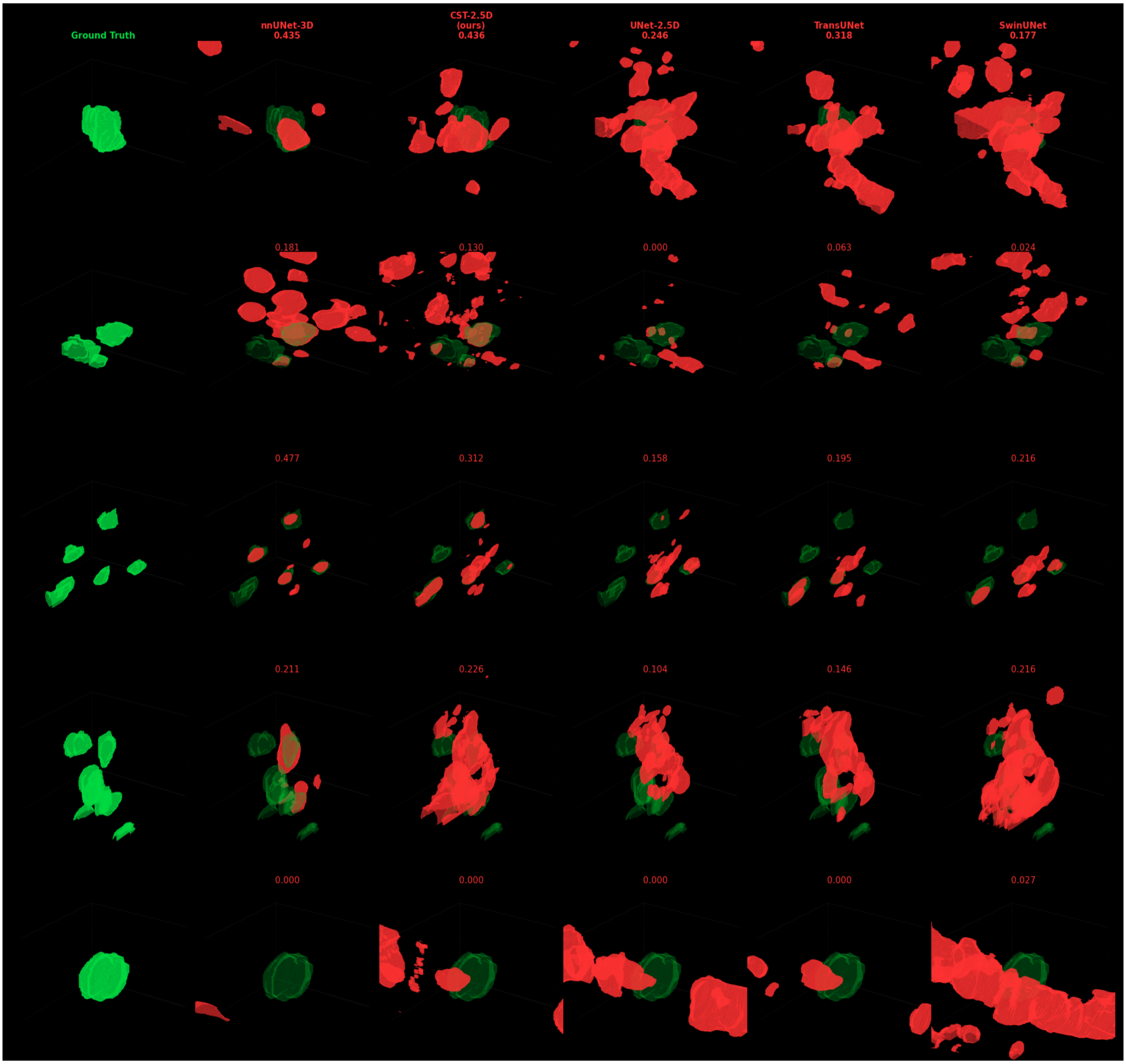

FIGURE 6. Representative failure cases on five Ver2 patients. GT: green; prediction: red; overlap: orange. Per-patient Dice shown above each column. Columns: Ground Truth, nnUNet-3D, CST-2.5D (ours), UNet-2.5D, TransUNet-2.5D, SwinUNet-2.5D. Failure patterns are analysed in Section V.

(7.97 M) and 2.5D design enable deployment on hardware that cannot support nnUNet-3D inference. However, the remaining gap to full 3D methods indicates that further development is required before clinical deployment. Essential prerequisites include interpretability tools (attention visualisation, uncertainty maps), PACS integration, and prospective clinical reader studies.

E. FUTURE DIRECTIONS

Future work should address the performance gap on small nodes through larger multi-institutional datasets, semi-supervised learning using the partially annotated patients in our corpus, and boundary-aware or distance transform loss terms. Federated learning across centres, self-supervised CT pretraining, and adaptive loss weighting to dynamically balance the precision-recall trade-off are also promising directions.

VI. CONCLUSION

We proposed the Cross-Slice Transformer (CST), a lightweight 2.5D architecture for CT lymph node segmentation that learns adaptive slice importance through statistical descriptor extraction and transformer-based attention. On the mediastinal-focused Ver2 dataset, CST achieves the highest

Dice among 2.5D models (0.5428 ± 0.0440), leading SwinUNet by 1.1 percentage points, and the highest node detection rate across all size strata: 53.6% for small nodes (<10 mm), 82.9% for medium nodes (10–15 mm), and 92.5% for large nodes (≥ 15 mm) at IoU ≥ 0.1. nnUNet-3D achieves higher overall Dice on both datasets, confirming that full volumetric context remains the strongest baseline when resources permit. Among 2.5D architectures, CST leads on node detection rate across all size strata on Ver2 and on small and medium nodes on Ver1, while achieving the highest recall (0.6179 on Ver2), surpassing both SwinUNet (0.6039) and nnUNet-3D (0.6032). Statistical testing confirms CST significantly outperforms all 2.5D baselines on Ver1 and TransUNet on Ver2 ($p \leq 0.004$), while the CST–SwinUNet difference on Ver2 is statistically equivalent ($p = 0.871$) with CST's advantage better reflected in detection rate and parameter efficiency. Higher recall is generally considered clinically preferable in lymph node staging where missed detections carry higher consequence than false alarms [5]. All models improve by 0.047–0.152 Dice points when trained on anatomically focused mediastinal data compared to multi-region data, demonstrating that dataset composition is a primary driver of segmentation performance across all architectural families. Absolute performance on small nodes remains moderate across all models, underscoring the need for larger annotated datasets, improved loss formulations, and rigorous multi-centre clinical validation before deployment in cancer staging workflows.

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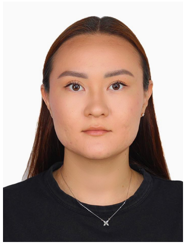

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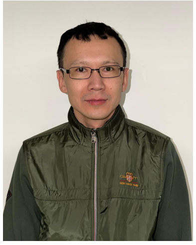

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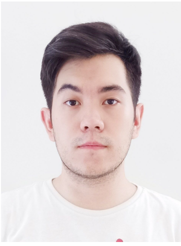

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